

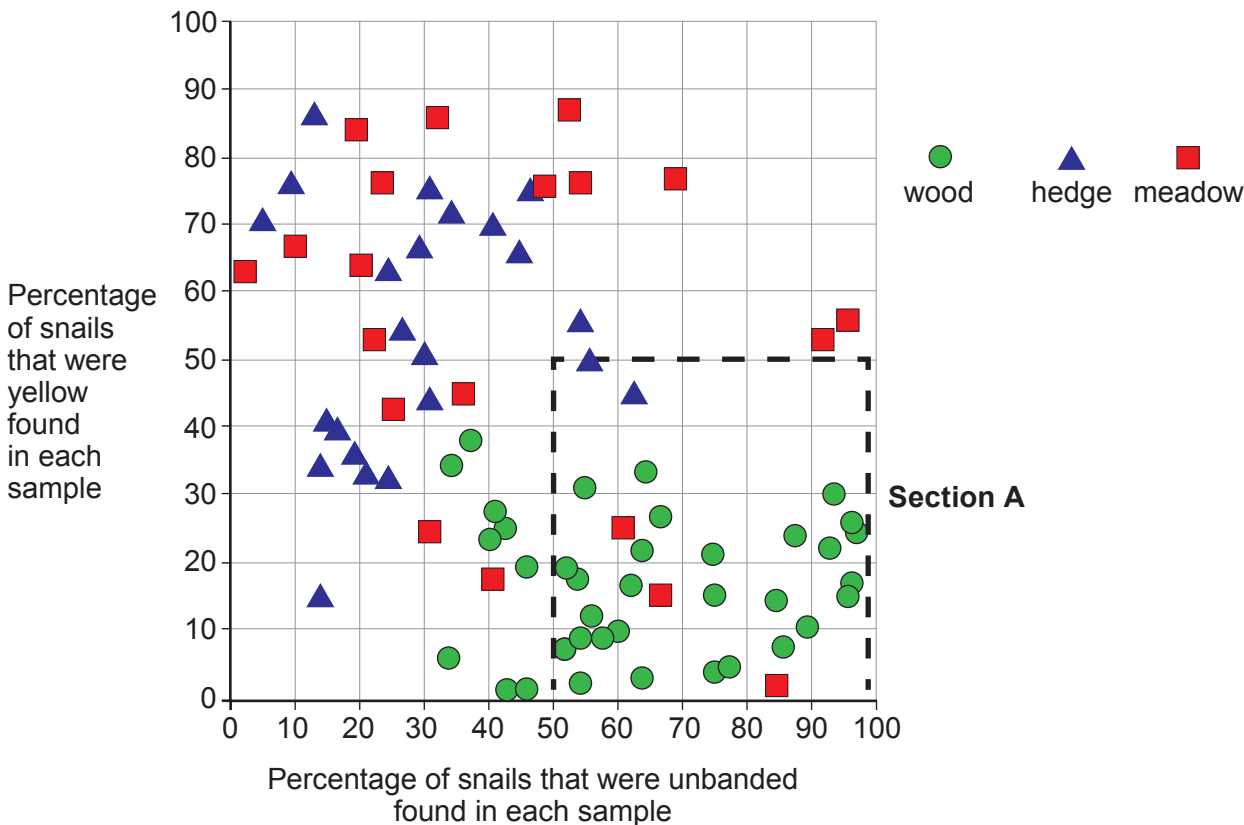
4. *Cepaea nemoralis* is a common British snail which is found in a variety of habitats. The shells of this species of snail vary in the pattern of dark bands found on their surface and their colours. They are either yellow or brown and banded or unbanded. The four phenotypes are shown in **Image 4.1**.

Image 4.1



An investigation was carried out to determine the phenotypes of the snails in a number of different habitats (woodland, hedge and meadow). Snails were collected from a number of different sample areas in each habitat and the percentage of each phenotype was determined. The results are shown in **Graph 4.2**. The different habitats are shown in the key. Each shape on the graph represents the results for one sample area.

Graph 4.2



Examiner
only

- (a) (i) Using data from **Graph 4.2**, state the conclusions you could draw about the distribution of the phenotypes in each of the **three** habitats. [3]

Wood

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Meadow

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Hedge

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- (ii) Explain why the snails might be distributed in the three habitats in such a way. [1]

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- (b) The snails collected from the sample areas shown in **Section A of Graph 4.2** were mainly brown and unbanded. The data suggests that there is an uneven distribution of this phenotype between the three habitats.

To statistically test this correlation, a Chi-squared (χ^2) statistical test was applied to the data. The null hypothesis was that ‘There is no statistically significant difference in the percentage of brown unbanded snails in the three different habitats’.

Calculate the value of Chi-squared (χ^2) by completing the table and using the formula below. [3]

Habitat	Observed (<i>O</i>)	Expected (<i>E</i>)	<i>O</i> – <i>E</i>	(<i>O</i> – <i>E</i>) ²	$\frac{(O - E)^2}{E}$
Wood	28				
Hedge	2				
Meadow	3				
	Total = 33				Σ =

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\chi^2 = \dots\dots\dots$$



Table 4.3

Chi-squared value

Degrees of freedom	Probability (p)									
	0.95	0.90	0.80	0.70	0.50	0.30	0.20	0.10	0.05	0.01
1	0.004	0.02	0.06	0.15	0.46	1.07	1.64	2.71	3.84	6.64
2	0.10	0.21	0.45	0.71	1.39	2.41	3.22	4.60	5.99	9.21
3	0.35	0.58	1.01	1.42	2.37	3.66	4.64	6.25	7.82	11.34
4	0.71	1.06	1.65	2.20	3.36	4.88	5.99	7.78	9.49	13.28

(c) **Table 4.3** shows a range of Chi-squared values. Using two degrees of freedom and the Chi-squared values shown in **Table 4.3**, state whether you would accept or reject the null hypothesis stated in (b). Explain your answer. [3]

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